



CLOUD COMPUTING

AND AWS

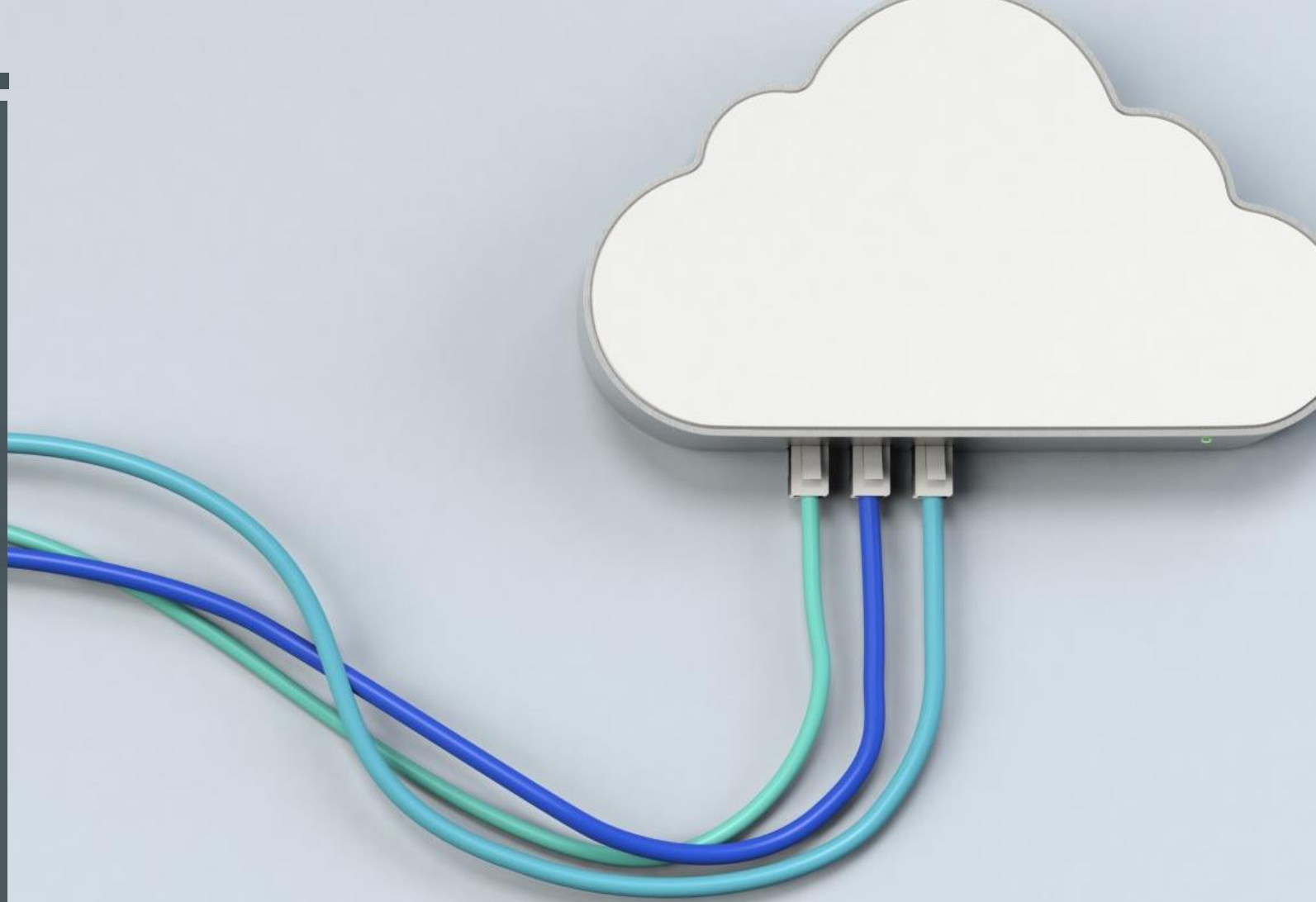


IN THIS PRESENTATION

- What is Cloud Computing?
- Cloud Computing Models
- Service Types
- What is Big Data?
- Architecture of Big Data
- Data Lifecycle

WHAT IS CLOUD COMPUTING?

- The 'cloud' refers to servers that are accessed over the internet and software and databases that run on those (remote) servers
- These servers are housed in data centers all over the world
- Thanks to cloud computing, companies and users do not need to manage physical servers themselves
- Allows us to back up and restore data from the cloud
- Improves collaboration, accessibility, mobility, and storage capacity
- Provides more advanced data security



CLOUD COMPUTING MODELS

- Cloud models describe where the cloud resources live
 - Public cloud – shared infrastructure(AWS,Azure, GCP, etc.)
 - Cost effective and scalable
 - Private cloud – dedicated to a single organization
 - Custom security and compliance
 - Hybrid cloud – mix of public and private
 - flexibility
 - Multi-cloud – multiple providers
 - Avoid vendor lock-in

SERVICE TYPES

- Service types describe how much control the user has
- Infrastructure as a Service (IaaS) - provides virtualized computing resources like servers and storage
 - Amazon EC2, Google Compute Engine(GCE), Azure
- Platform as a Service (PaaS) - offers ready-to-use development and deployment environment without managing the underlying infrastructure
 - AWS elastic beanstalk, Google App Engine, Heroku
- Software as a Service (SaaS) - delivers fully functional software applications over the internet
 - Microsoft 365, Google Workspace, Salesforce

WHAT IS BIG DATA?

- Big Data refers to the vast and rapidly growing volumes of data that are too large and complex for traditional data processing tools to manage
- This data comes in many forms:
 - Structured – tables
 - Semi-structured – JSON or XML
 - Unstructured – text, images, video
- Organizations use big data to improve operational efficiency, enhance risk management, innovate faster, and predict customer behavior

5 V'S OF BIG DATA

- **Volume** – amount of data
- **Velocity** – speed of data generation, processing, and sharing
- **Variety** – data formats, structured & unstructured
- **Veracity** – data quality
- **Value** – insights gained on relevant data

ARCHITECTURE OF BIG DATA

- The architecture of big data is a pipeline that handles data ingestion, storage, processing, analytics, and visualizations
- The exact architecture can vary greatly from project to project
- Roughly follows a 3-step process
 - Data integration – collect data from multiple sources
 - Storage/management - store data in lakes or distributed file systems
 - Analysis/visualization - run analytics and extract insights



DATA LIFECYCLE

- **Data Creation / Collection**

- Data is generated from sources
- Applications, sensors, transactions, users

- **Data Ingestion**

- Data is moved into storage systems
- Batch or real-time ingestion

- **Data Storage**

- Data is stored in databases or data lakes
- Raw and processed data retained

- **Data Processing**

- Data is cleaned, transformed, aggregated
- Batch or streaming processing

- **Data Analysis**

- Data is queried and analyzed
- Used for reporting, analytics, and ML

- **Data Visualization & Consumption**

- Insights delivered to users or systems
- Dashboards, reports, APIs

- **Data Archival / Deletion**

- Old data archived or removed